

Implementing a calculator: question

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Learning outcomes

- A Motivating exercise: “Write an interpreter ...”

A motivating example

- We have (at least) the following numerical systems:
 - a Haskell built-in `Int` or `Integer`
 - Binary numerical systems: `BN`, `BNList`, `BNEither`
- Define a language for numerical expressions built on numerical constants, Additions, Multiplications:

```
-- ex using BN
(XH `+` (XO XH)) `*` (XI XH)
-- ex using BN
([] `+` [False]) `*` [True]
```

- Define an interpreter from the language of expressions to at least one numerical system:

```
-- ex using BN
eval ((XH `+` (XO XH)) `*` (XI XH)) == XI (XO (XO XH))
-- ex using BN
eval (([] `+` [False]) `*` [True]) == [True, False, False]
```